

GHW_1

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a	b	$a \vee b$	$a \sqcup b$	$a \sqcup (a \sqcup b)$
T	T	T	T	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	F

We will define the squand operator " $a \sqcup b$ " as equivalent to the proposition $a \vee \neg b$. From the proposition $a \vee b$ we can expand to $(a \vee b) \wedge T$ or $(a \vee b) \wedge (a \vee \neg a)$. Then using the distributive property we can pull the a element out $a \vee (\neg a \wedge b)$. Then using DeMorgan's law we get $a \vee \neg(\neg a \vee \neg b)$. Plugging in the squand identity we first get $a \vee \neg(a \sqcup b)$ and then $a \sqcup (a \sqcup b)$.